

Tool /framework assumptions — status delta

QDS assumptions_report grew from 33 → 36+ rows · two upgrades, four hardenings, two new mitigations

Assumption	Was	Now
ASSUM_interp_ion_identity_no_anomalous	known_violation	mitigated (CheckMyMetal)
ASSUM_phenix_refine_in_run_r_factors_use_internal_scaling	known_violation	known_violation (3-tool hardened)
ASSUM_find_peaks_holes_cutoff_and_merging	known_violation	
ASSUM_report_round_table_collapses_rounds	known_violation	
ASSUM_report_position_initial_not_final	known_violation	
ASSUM_report_water_count_stage_drift	known_violation	
ASSUM_agg_zero_rama_rounds_one_outlier_to_zero	mitigated	mitigated (3-tool Asn A 39 hardens)
ASSUM_checkmymetal_geometry_based	—	unchecked → mitigated by 1SAR run
ASSUM_t13_oracle_pending (new)	open follow-up	closed — ctruncate wired
ASSUM_aimless_requires_unmerged_intensities (new)	—	mitigated — wrapper falls back to ctruncate

Open assumption — K⁺ alternative

K⁺ alternative for both ions has the best raw bond-length fit but coordination is too low.
Will be added as an "unchecked" assumption if buffer content is investigated.

Trust model preserved: every load-bearing finding has ≥ 1 non-cctbx confirmation; 5 most consequential have ≥ 3.

Five findings with ≥ 3 confirmations: R-free · Asn A 39 · RMSD bracket · ligand RSCC · Mg²⁺ exclusion.